

# Common Rehabilitation Conditions HHS4185 復康科常見疾病 HHS4185J

**L3: Orthopaedic – Soft Tissue Problems: Acute Sprain and Repetitive Stress Injury**

**L3: 常見骨科疾病 – 軟組織問題: 扭傷, 重複性勞損**

Lecturer:  
講師:

Ms. Fion YEUNG (HHS4185)  
Ms. Shan LI (HHS4185J)

# Soft Tissues Overview

## 軟組織概述

# What is “soft” tissue? 什麼是「軟」組織？

An umbrella term for many forms of non-hardened tissue in body

- **Hard tissues**
  - Bones
- **Soft tissues**
  - Skin
  - Subcutaneous fat
  - Fascia
  - Muscles
  - Ligaments
  - Tendons

一個涵蓋身體多種非硬組織的總稱

- **硬組織**
- 骨骼
- **軟組織**
- 皮膚
- 皮下脂肪
- 筋膜
- 肌肉
- 韌帶
- 腱

# Ligaments 韌帶

Connective tissues that connect bones together

- Made up of bundles of collagen fibers
- Attached to outer covering of bones – periosteum
- Blood supply – at insertion site

## Functions:

- Restrict joint motion to prevent excessive joint movements)
- Stabilize joint
- Joint proprioception

## 連接骨骼的結締組織

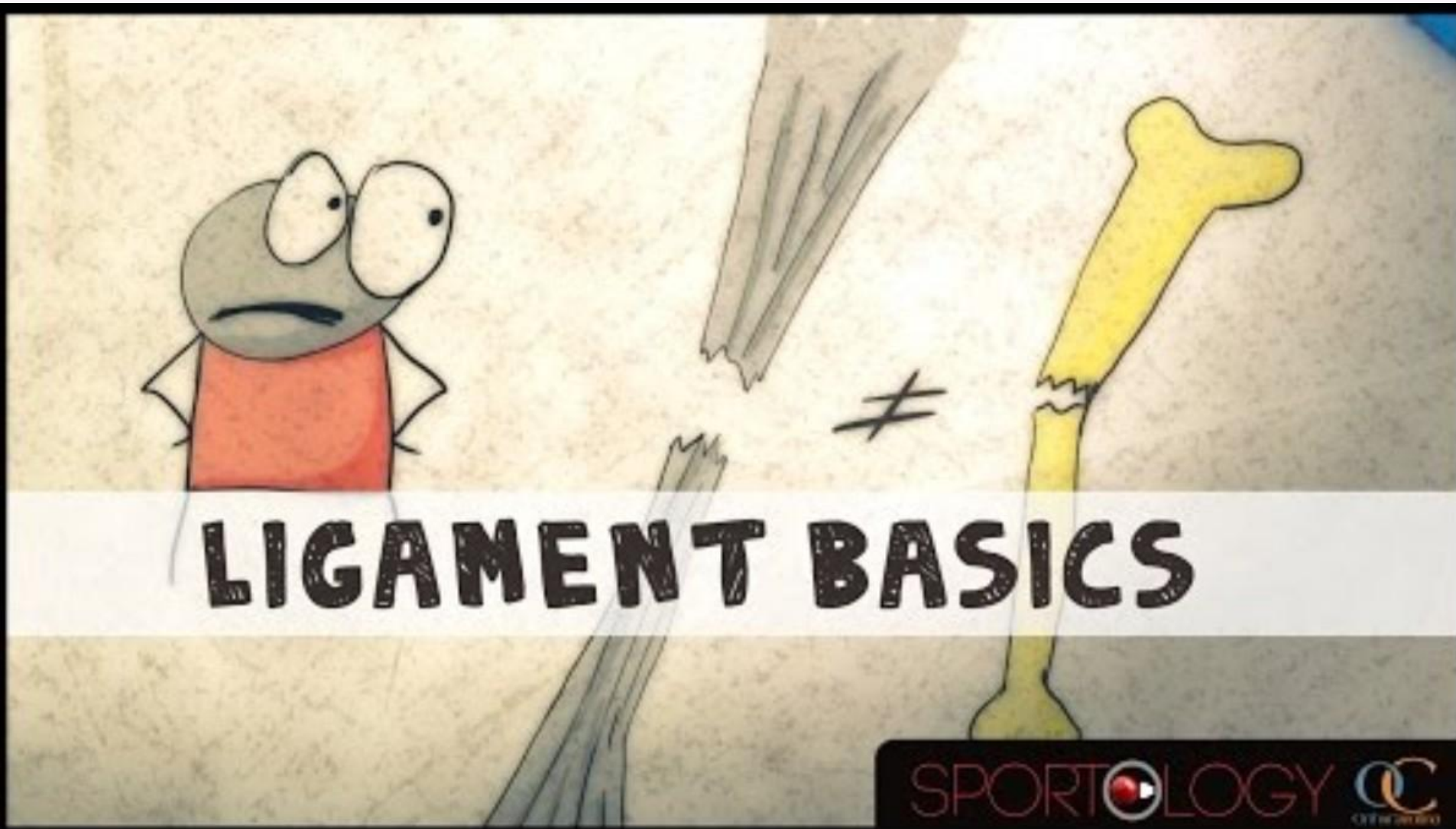
- 由膠原纖維束組成
- 附著於骨骼的外覆層—骨膜
- 血液供應—在附著部位

## 功能：

- 限制關節運動以防止過度的關節移動
- 穩定關節
- 關節本體感覺

## Ligament Basics – Science Explained:

<https://youtu.be/V3ZGonL8QgM>



# Tendons 筋

Dense connective tissues that connect muscles to bones

- Made up of bundles of collagen fibers
- Each muscle has 2 tendons: proximal & distal
- Blood supply – musculotendinous junction, osseotendinous junction

## Functions:

- Transfer forces from muscles to bones to produce joint motion

密集結締組織，連接肌肉與骨骼

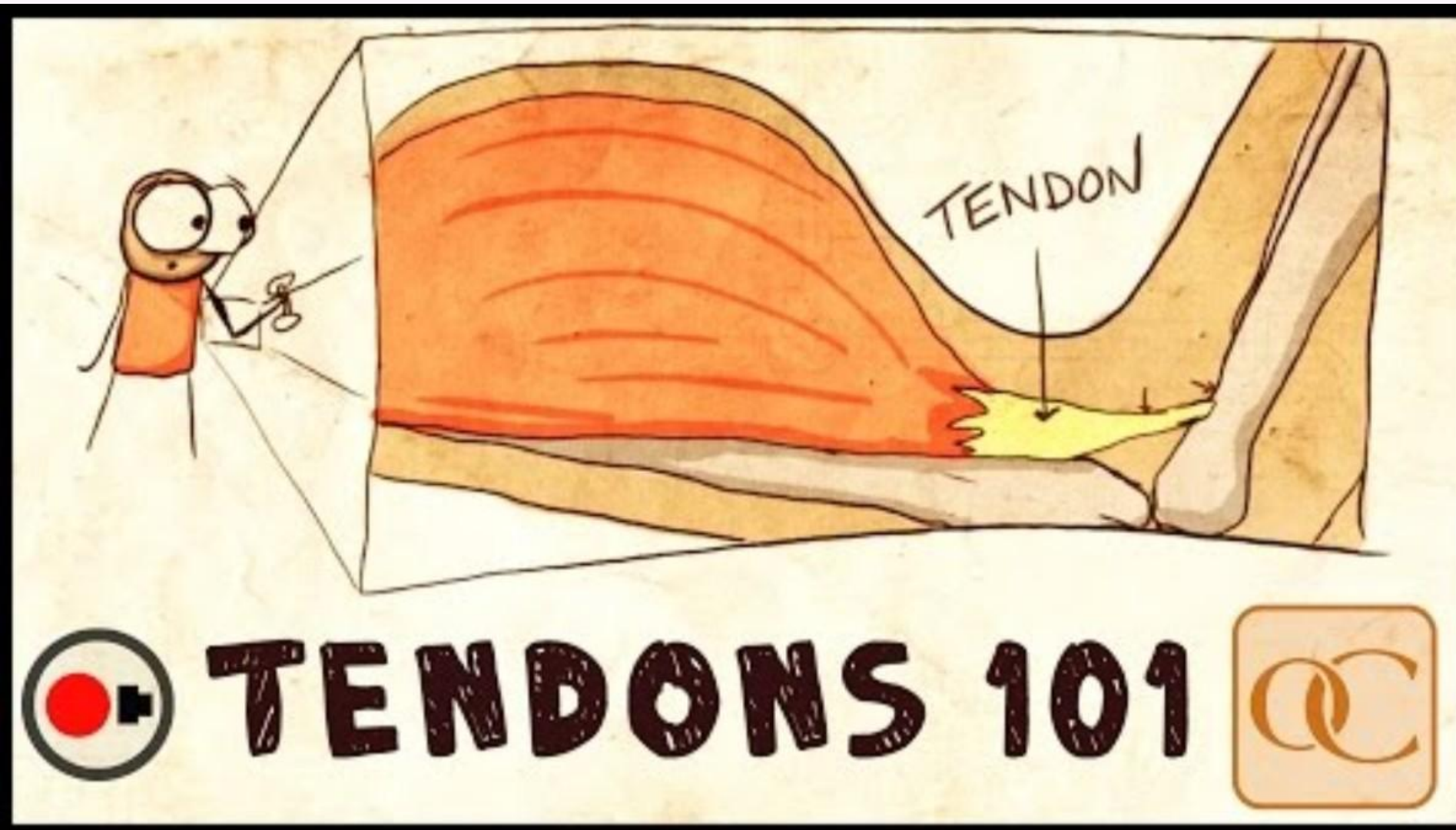
- 由膠原纖維束組成  
每個肌肉有兩條腱：近端腱和遠端腱
- 血液供應—肌腱交界處、骨腱交界處

## 功能：

將力量從肌肉轉移到骨骼，以產生關節運動



The Basic Science of Tendons & Tendinitis:  
<https://www.youtube.com/watch?v=hdes6W7600w>



# Skeletal Muscles 骨骼肌

## Skeletal muscles attach to bones via tendons

- Made up of muscle fibres, wrapped in a dense connect tissue sheath
- Blood supply – branches of arteries and capillary network penetrate in muscles

## Classification:

- Type I – slow-twitch fibres contract slowly, moderate force level, resist fatigue
- Type II – fast twitch fibres contract rapidly, high force level, fatigue quickly

## Functions:

- Voluntary movements
- Posture & stability
- Storage of protein
- Heat production
- Joint protection

## 骨骼肌通過腱附著於骨骼

- 由肌纖維組成，包裹在密集的結締組織鞘中
- 血液供應—動脈分支和毛細血管網絡深入肌肉

## 分類：

- I型—慢縮纖維緩慢收縮，產生中等力量，抗疲勞
- II型—快縮纖維迅速收縮，產生高力量，迅速疲勞

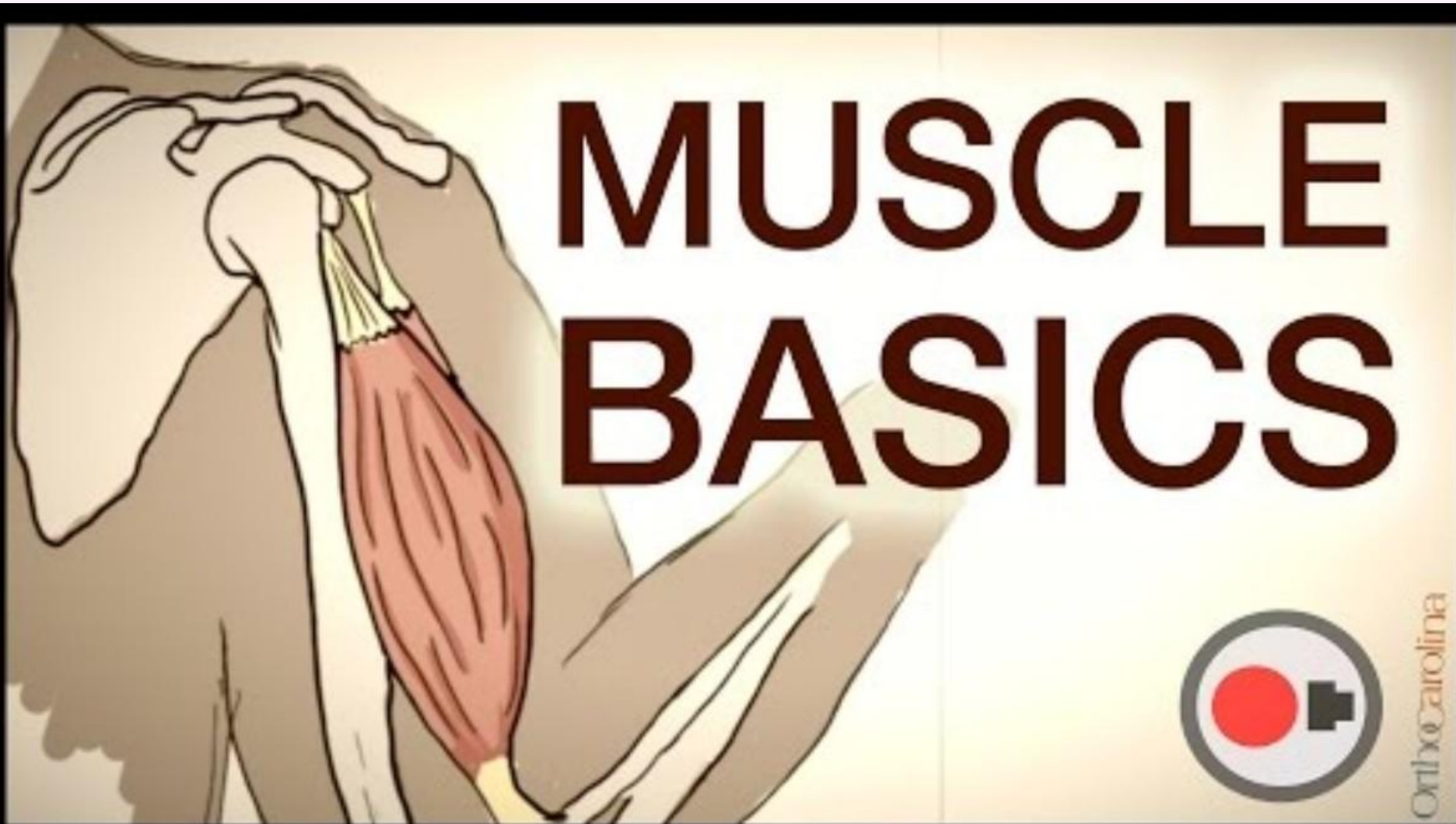
## 功能：

- 自願運動
- 姿勢與穩定性
- 蛋白質儲存
- 產熱
- 關節保護



## Muscle Basics: What Athletes Need to Know About the Muscular System:

<https://www.youtube.com/watch?v=94Q-fvCAJzQ>



# Soft Tissue Injuries

## 軟組織損傷



# Soft Tissue Injury 軟組織損傷

Injury happens in the “soft” tissues of the musculoskeletal system

- Muscles
- Tendons
- Ligaments

Types of soft tissue injuries:

- Acute (immediate)
- Repetitive strain

損傷發生在肌肉骨骼系統的“軟”組織中

- 肌肉
- 腱
- 韌帶

軟組織損傷的類型：

- 急性（即時）
- 重複性損傷

# Acute (immediate) soft tissue injuries

## 急性（即時）軟組織損傷

Usually results of a single traumatic event / accident

### Strains:

- Overstretch / tear a muscle or tendon
- Sudden / jerky movement

### Sprains:

- Overstretch / tear a ligament
- Sudden / jerky movement

### Contusions:

- A bruise with broken blood vessels causing discolouration under the skin
- Forceful impact from a direct blow, fall or collision

通常是單次外傷事件/事故的結果  
拉傷：

- 過度伸展/撕裂肌肉或腱
- 突然/急促的動作

扭傷：

- 過度伸展/撕裂韌帶
- 突然/急促的動作

挫傷：

- 皮下血管破裂造成的瘀傷，導致皮膚變色
- 由直接撞擊、跌倒或碰撞造成的強烈衝擊

# Repetitive Strain Injuries 重複性勞損

**Causes by repetitive motion / overuse despite low loading**

## **Tendonitis:**

- Inflammation & swelling in tendon caused by overuse / repetitive motion
- Microscopic tears in tendon fibres

## **Chronic muscle strains:**

- Muscle injuries caused by overuse / repetitive motion

**由於重複動作/過度使用，即使負荷較低而引起**

## **腱炎：**

- 由於過度使用/重複動作引起的腱部炎症和腫脹
- 腱纖維中的微小撕裂

## **慢性肌肉拉傷：**

由於過度使用/重複動作引起的肌肉損傷



# Soft Tissue Healing 軟組織癒合

## Timeline of healing depends on:

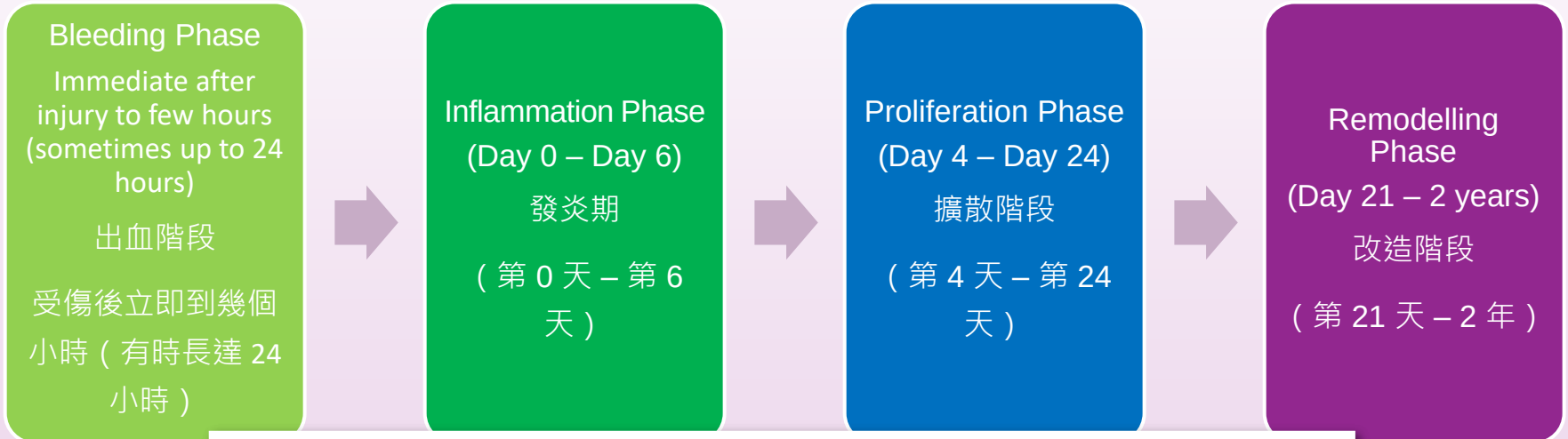
- Individual health status
- Age
- Extent of injury
- Injury tissue type
- Injury site

## 癒合的時間表取決於：

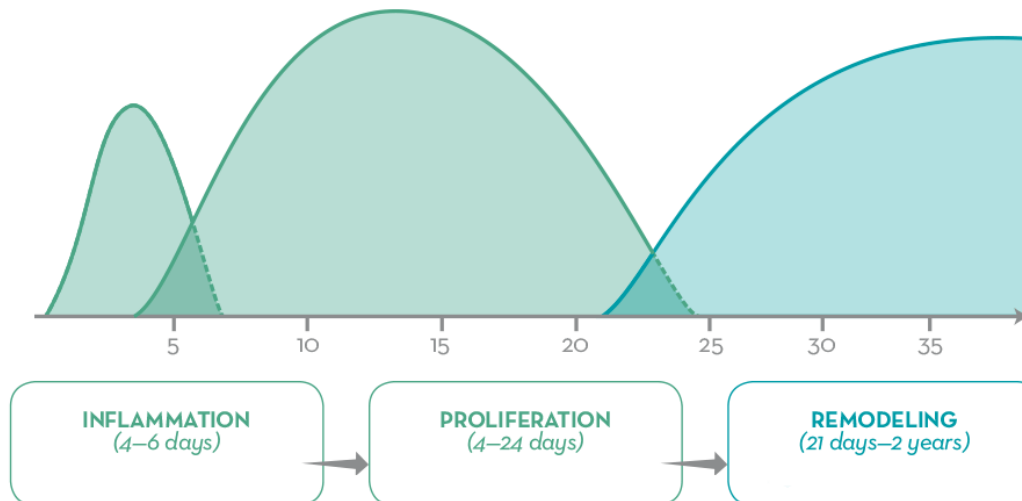
- 個體健康狀況
- 年齡
- 受傷程度
- 受傷組織類型
- 受傷部位

# Phases of Soft Tissue Healing

## 軟組織癒合的階段



### STAGES OF WOUND HEALING



**TABLE 2** Approximate rates of tissue healing<sup>a</sup>

| Tissue and grades of injury | 0–3 d | 4–14 d | 3–4 wk | 5–7 wk | 2–3 mo | 3–6 mo | 6–12 mo | >1 year                |
|-----------------------------|-------|--------|--------|--------|--------|--------|---------|------------------------|
| Skin                        |       |        |        |        |        |        |         |                        |
| SQ                          |       |        |        |        |        |        |         |                        |
| Fascia                      |       |        |        |        |        |        |         |                        |
| Muscle                      |       |        |        |        |        |        |         |                        |
| DOMS (exercise induced)     |       |        |        |        |        |        |         |                        |
| Grade 1                     |       |        |        |        |        |        |         |                        |
| Grade 2                     |       |        |        |        |        |        |         |                        |
| Grade 3                     |       |        |        |        |        |        |         |                        |
| Tendon                      |       |        |        |        |        |        |         |                        |
| Acute                       |       |        |        |        |        |        |         |                        |
| Subacute                    |       |        |        |        |        |        |         |                        |
| Chronic                     |       |        |        |        |        |        |         |                        |
| Rupture/surgical repair     |       |        |        |        |        |        |         |                        |
| Ligament (extra-articular)  |       |        |        |        |        |        |         |                        |
| Grade 1                     |       |        |        |        |        |        |         |                        |
| Grade 2                     |       |        |        |        |        |        |         |                        |
| Grade 3                     |       |        |        |        |        |        |         |                        |
| Intra-articular             |       |        |        |        |        |        |         | Unlikely to fully heal |
| Bone                        |       |        |        |        |        |        |         |                        |

Abbreviations: DOMS, delayed onset muscle soreness; SQ, subcutaneous.

<sup>a</sup>Expected time frame for tissue healing after injury. Rate of healing is influenced by the degree of tissue damage (Grade), particularly with muscle, tendon, and ligament injury. Muscle: Grade 1, mild damage (<5% of fibers), minimal loss of strength and function; Grade 2, moderate fiber damage, loss of strength and function; Grade 3, complete rupture of muscle/muscle-tendon and loss of function. Ligament: Grade 1, stretching, little/no tear, no joint instability; Grade 2, partial tear, mild instability; Grade 3, complete rupture, loss of function. The shaded cells correspond to the range of healing time for the specific tissue injury indicated in the left column. Healing time varies based on degree of tissue injury.

# Complications of Soft Tissue Injuries

## 軟組織損傷的並發症

**Acute complications occur with soft tissue injuries:**

**Blood vessels damage**

- Haemorrhage - serious internal bleeding
- Ischaemia - loss of blood supply to body tissues

**Nerves damage**

- Direct trauma
- Compression due to swelling

**Compartment syndrome**

- Severe swelling within muscle compartment (fascia)
- Intense pain and cut off blood supply

**急性併發症發生在軟組織損傷中：**

**血管損傷**

- 出血 - 嚴重內部出血
- 缺血 - 體組織失去血液供應

**神經損傷**

- 直接創傷
- 由於腫脹造成的壓迫

**腔室綜合症**

- 肌肉腔室（筋膜）內嚴重腫脹
- 劇烈疼痛並切斷血液供應

# Complications of Soft Tissue Injuries

## 軟組織損傷的並發症

### Chronic complications due to improper healing:

#### Contracture

- Immobilization during healing phase

#### Joint instability

- Severe ligament sprain with incomplete healing

#### Chronic inflammation

- Incomplete healing from repetitive strain injuries with chronic pain and long-term tissue damage

#### Heterotopic Ossification

- Scarring with bone tissue develops within soft tissue

### 由於不當癒合引起的慢性併發症：

#### 挛縮

- 癒合階段的固定

#### 關節不穩定

- 嚴重韌帶扭傷未完全癒合
- 慢性炎症

### 由於重複性損傷導致的

- 完全癒合，伴隨慢性疼痛和長期組織損傷

#### 異位骨化

- 軟組織內發展出含骨組織的疤痕

# Treatment Principle 治療原理 - POLICE

Traditional principles – RICE  
Rest, Ice, Compress, Elevate

Modern principles – POLICE

- P = Protection
- OL = Optimal Loading
- I = Ice
- C = Compression
- E = Elevation

傳統原則 – RICE  
休息、冰敷、壓迫、抬高

現代原則 – POLICE

- P = 保護
- OL = 最佳負荷
- I = 冰敷
- C = 壓迫
- E = 抬高



# POLICE – P for Protection

## P 代表保護

### Objective:

- To prevent further damage to the injured area

### Action:

- Protect injured part from further injury
- Using supports like splints, braces, or crutches to avoid movements that could worsen the injury

### Why Important:

- Limits additional damage
- Stabilizes the injury
- Prevent further trauma other soft tissues

### 目標：

- 防止受傷區域進一步損傷

### 行動：

- 保護受傷部位免受進一步傷害
- 使用支撐物，如夾板、護具或拐杖，避免可能加重傷勢的動作

### 為什麼重要：

- 限制額外損傷
- 穩定傷勢
- 防止其他軟組織進一步受創

# POLICE – OL for Optimal Loading

## OL 實現最佳負載

### Objective:

- To accelerate healing process as body tissues require some loading to start healing

### Action:

- Gradual introduction of activity & movement under proper protection
- Gentle manual techniques, exercise with supports

### Why Important:

- Right amount of activity help to manage swelling with muscle pumping
- Prolonged rest / immobilization may lead to complications such as contracture

### 目標：

- 加速癒合過程，因為身體組織需要一定的負荷來開始癒合

### 行動：

- 在適當保護下逐步引入活動和運動
- 輕柔的手法、使用支撐進行運動

### 為什麼重要：

- 適當的活動量有助於通過肌肉抽動來控制腫脹
- 長時間的休息/固定可能導致如挛縮等併發症

# POLICE – I for Ice

## I代表冰

### Objective:

- To reduce acute inflammation, pain & swelling

### Action:

- Apply ice / cold pack for 15-20 min every 1-2 hours during first 48-72 hours after injury
- Ice until resting pain subsided

### Why Important:

- Constricts blood vessels, reduce blood flow to injured area, limits swelling & bruising
- Immediate pain relief with pain gate theory

### 目標：

- 減少急性炎症、疼痛和腫脹

### 行動：

- 在受傷後的前48-72小時內，每1-2小時使用冰袋/冷敷15-20分鐘
- 冰敷直到靜止時的疼痛減輕

### 為什麼重要：

- 收縮血管，減少受傷區域的血流，限制腫脹和瘀傷
- 根據疼痛門理論，立即緩解疼痛

# POLICE – C for Compression

## C 代表壓縮

### Objective:

- To minimize swelling
- To provide support to injured area

### Action:

- Apply elastic bandage or compression wrap to gently compress injured area
- Avoid tight compression that restrict circulation

### Why Important:

- Reduce fluid build up in the tissues
- Reduce swelling

### 目標：

- 最小化腫脹
- 為受傷區域提供支撐

### 行動：

- 使用彈性繃帶或壓縮包裝輕柔地壓迫受傷區域
- 避免過緊的壓迫以免限制血液循環

### 為什麼重要：

- 減少組織內的液體積聚
- 減少腫脹

# POLICE – E for Elevation

## E 代表抬高

### Objective:

- To reduce swelling
- To promote drainage of excess fluid

### Action:

- Elevate the injured body part to above heart level for 20-30 min, several times a day, esp. first 48 hours

### Why Important:

- Encourages venous return
- Reduce swelling

### 目標：

- 減少腫脹
- 促進多餘液體的排出

### 行動：

- 將受傷部位抬高至心臟以上，持續20-30分鐘，每天多次，特別是在前48小時內

### 為什麼重要：

- 促進靜脈回流
- 減少腫脹

# Ligament Sprain 扭傷

# Ligament Sprain 韌帶扭傷

- Common injuries
- When ligament is over-stretched or torn due to excess range of movement
- Common injury area:
  - Ankle
  - Knee
  - Wrist

- 常見傷害
- 當韌帶因過度活動範圍而過度拉伸或撕裂時
- 常見受傷部位：  
腳踝  
膝蓋  
手腕



# Ligament Sprain Injury Grading

## 韌帶扭傷分級

### Grade I

- Structural damage only on microscopic level
- Slight local tenderness
- No joint instability

### Grade II

- Partial tear of the ligament
- Visible swelling
- More tenderness
- No joint instability -> mild instability

### Grade III

- Severe sprain with complete tear
- Significant swelling
- Joint instability

### 級別 I

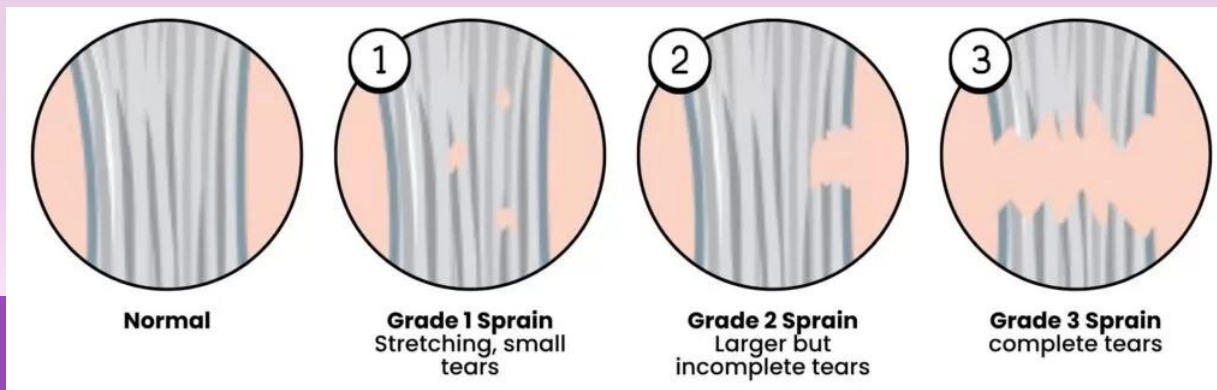
- 結構損傷僅在顯微層面
- 輕微局部觸痛
- 無關節不穩定

### 級別 II

- 韌帶部分撕裂
- 可見腫脹
- 觸痛加重
- 無關節不穩定 → 輕微不穩定

### 級別 III

- 嚴重扭傷，完全撕裂
- 顯著腫脹
- 關節不穩定



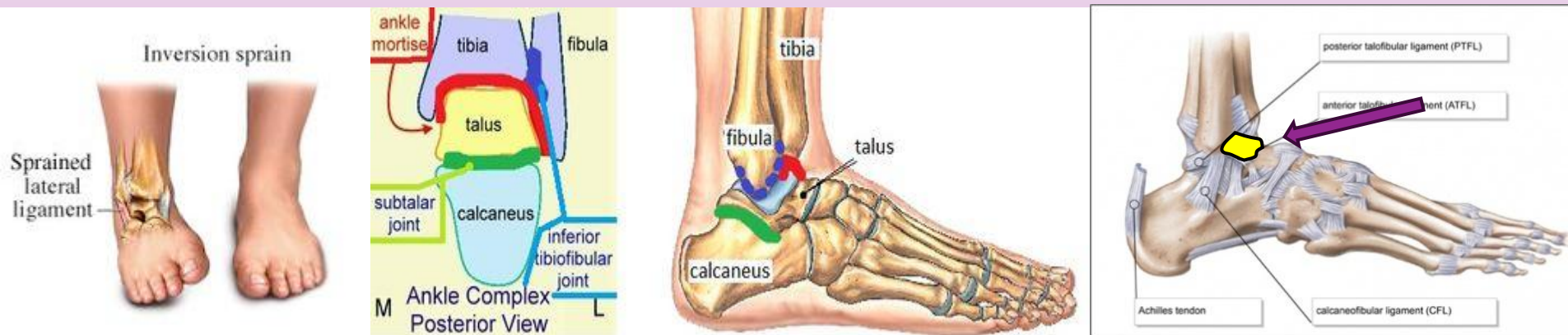
# Lateral Ankle Sprain 踝關節外側扭傷

Ankle joint made up with the tibia, fibular and talus

- Occurs when the foot rolled inward (ankle inversion + plantarflexion)
- Lateral Ankle Sprain most commonly involves the Anterior Talo-Fibular Ligament (ATFL) in the talocrural joint (ankle joint)

踝關節由脛骨、腓骨和距骨組成

- 當腳向內翻轉（踝部內翻 + 足底屈曲）時發生
- 外側踝扭傷最常涉及距腓前韌帶（ATFL），位於距骨關節（踝關節）中



# Lateral Ankle Sprain 踝關節外側扭傷

- Most common lower limb musculoskeletal injuries for people participate in recreational physical activities
- Most common type of ankle sprains – around 85% of all ankle sprains
- Cause:
  - Fall
  - Sudden change of ankle movement
  - Being hit by hard object
  - Wearing shoes do not fit
  - Walking on uneven surface
  - Sport activities

- 參加休閒體育活動的
- 最常見的下肢肌肉骨骼傷害
- 最常見的踝扭傷類型 – 約佔所有踝扭傷的85%

原因：

- 跌倒
- 踝部運動的突然變化
- 被硬物撞擊
- 穿著不合適的鞋子
- 在不平坦的地面上行走
- 體育活動

[醫生講場] 常見運動創傷 – 腳踝篇及腳踝復康運動  
<https://www.youtube.com/watch?v=yMsVe4HECxs>



# Lateral Ankle Sprain 踝關節外側扭傷

## Signs & symptoms:

- Pain when weightbearing on injured foot
- Tenderness on palpation
- Swelling
- Decreased range of motion
- Instability of ankle joint
- Inversion injury

## 徵兆與症狀：

- 在受傷腳上承重時疼痛
- 觸診時有壓痛
- 腫脹
- 活動範圍減少
- 踝關節不穩定
- 內翻傷害



# Diagnosis 診斷

- Subjective examination - onset, pattern, means to relieve
  - Objective examination:
    - Acute inflammation:
      - Redness, joint swelling, warmth & pain
    - Joint crepitus: clicking / popping
    - “Locking” / “sticking” during movement
    - Range of motion (active, assisted, passive)
    - Joint instability with muscle weakness / buckling
  - Imaging test: X-ray
- 問診 – 發病時間、症狀表現模式、減輕症狀的方法
  - 客觀檢查:
    - 急性炎症
      - 急性發炎 發紅、關節腫脹、發熱和疼痛
    - 關節磨擦音: 咔噠聲/噼啪聲
    - 關節活動時 “鎖” / “粘”
    - 關節運動範圍 (主動, 輔助, 被動)
    - 關節不穩定伴肌肉無力 / 屈曲
  - 影像學檢查: x光



# Lateral Ankle Sprain 踝關節外側扭傷

## Physiotherapy Management:

- Follow POLICE protocol in inflammatory phase (Day 0 – Day 6)
  - Aircast / orthopaedic boot
  - Elbow crutches if necessary, at least partial weight bearing (PWB)
- Mobilization exercise for ROM
- Strengthening exercise for muscle strength
- Proprioception training
- Functional activity training

## 物理治療管理：

- 在炎症期（第0天至第6天）
- 遵循POLICE原則
  - 使用氣墊護具/矯形靴
  - 如有必要，使用肘拐杖至少進行部分負重（PWB）
- 進行活動範圍（ROM）的動員運動
- 進行肌肉力量的加強運動
- 進行本體感覺訓練
- 進行功能性活動訓練

# Lateral Ankle Sprain

## Medical Management:

- Drug treatment: NSAIDs, non-opioid analgesics

## Surgical treatment:

- Only if conservative treatments fail
- Ankle Arthroscopy



## 醫療管理：

- 藥物治療：非類固醇消炎藥（NSAIDs）、非鴉片類鎮痛劑

## 手術治療：

- 僅在保守療法失敗時進行  
踝關節鏡檢查





# *Tendon / Muscle Strain*

## 肌腱 / 肌肉拉傷

# Tendon / Muscle Strain

## 肌腱/肌肉拉傷

- Common injuries
- When tendon / muscle is over-stretched or torn when the muscle is activated
- Damage mostly localized near muscle-tendon junction
- Common injury area:
  - Back muscles
  - Hamstrings muscles
  - Calves muscles
  - Groin, esp. adductors
  - Shoulders, esp rotator cuff
- 常見傷害
- 當肌腱或肌肉在激活時過度拉伸或撕裂
- 損傷主要集中在肌肉-肌腱交接處附近
- 常見受傷部位：
  - 背部肌肉
  - 腿後肌肉
  - 小腿肌肉
  - 腹股溝，特別是內收肌
  - 肩部，特別是旋轉袖

# Muscle Strain Injury Grading 肌肉拉傷分級

## Grade I (Mild)

- Affected limited no. of muscle fibres
- No decrease in strength, full active & passive ROM
- Pain & tenderness delayed to next day

## Grade II (Moderate)

- Nearly half of muscle fibres torn
- Swelling
- Minor decrease in strength
- Acute & significant pain

## Grade III (Severe)

- Complete rupture of muscle
- Complete loss of muscle function
- Severe swelling and pain

## 級別 I (輕度)

- 受影響的肌肉纖維數量有限
- 無力量下降，主動和被動活動範圍完整
- 疼痛和觸痛延遲至次日

## 級別 II (中度)

- 近一半的肌肉纖維撕裂
- 腫脹
- 輕微力量下降
- 急性且明顯的疼痛

## 級別 III (重度)

- 肌肉完全斷裂
- 肌肉功能完全喪失
- 嚴重腫脹和疼痛



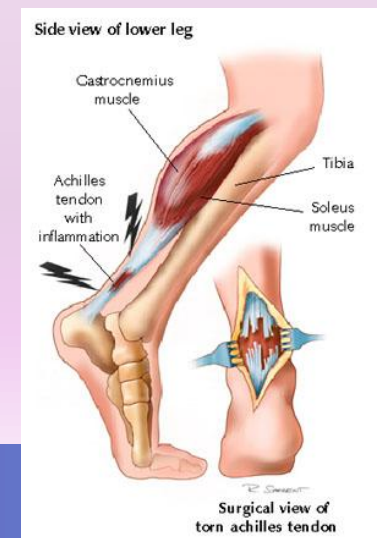
# Achilles Tendon Injury

## 跟腱損傷

- Achilles tendon is the strongest tendon in human body
- Connecting the calf muscles to heel bone
  - Calf muscles are major plantarflexors, including gastrocnemius and soleus
- Usually traumatic injury during sports event:
  - Sudden forced plantar flexion
  - Violent dorsiflexion in a plantar flexed foot



- 跟腱是人體中最強的肌腱
- 將小腿肌肉連接到跟
  - 小腿肌肉是主要的足底屈肌，包括腓腸肌和比目魚肌
- 通常在運動中發生創傷性損傷：
  - 突然的強迫足底屈曲在足底屈曲的腳上進
  - 劇烈的背屈



# Achilles Tendon Injury 跟腱損傷

- Common among “middle-aged weekend warriors” – 70% of this injury
- Common for “stop-and-go” sports such as badminton, soccer, volleyball, basketball, tennis, squash, etc.
- Causes:
  - Fall from height
  - Forceful pushing off with a weight bearing forefoot
  - History of a prior cortisone injection
- 在「中年週末戰士」中常見 – 70% 的此類傷
- 在「停停走走」的運動中常見，如羽毛球、足球、排球、籃球、網球、壁球等
- 原因：
  - 從高處跌落
  - 用承重的前腳用力推起
  - 有過去使用皮質類固醇注射的歷史



# Sick問識答 | 做運動熱身不足易「斷腳」 拆解阿基里斯腱斷裂成因 + 3大治療方案

<https://topick.hket.com/article/3845681>

Sick問  
識答

睇片



示意圖



骨科醫生・楊旭楠醫生



Topick  
hket.com

## 做運動熱身不足易「斷腳」 拆解阿基里斯腱斷裂成因 + 3大治療方案

# Achilles Tendon Injury 跟腱損傷

## Signs & symptoms:

- Sharp pain at the time of rupture
- Loud crack or pop sound when rupture
- A gap may be felt when palpate on tendon
- Swelling of back of the heel
- Decrease active plantarflexion
- Increase passive dorsiflexion
- Decrease muscle strength
- Impaired walking gait

## 徵兆與症狀：

- 在撕裂時有尖銳的疼痛
- 撕裂時聽到響亮的裂響或爆裂聲
- 觸診時可能感覺到肌腱
- 凹陷
- 跟部後方腫脹
- 主動足底屈曲減少
- 被動背屈增加
- 肌肉力量下降
- 行走步態受損



# Diagnosis 診斷

- Subjective examination - onset, pattern, means to relieve
  - Objective examination:
    - Acute inflammation:
      - Redness, joint swelling, warmth & pain
    - Joint crepitus: clicking / popping
    - “Locking” / “sticking” during movement
    - Range of motion (active, assisted, passive)
    - Joint instability with muscle weakness / buckling
  - Imaging test: X-ray
- 問診 – 發病時間、症狀表現模式、減輕症狀的方法
  - 客觀檢查:
    - 急性炎症
      - 急性發炎 發紅、關節腫脹、發熱和疼痛
    - 關節磨擦音: 咔噠聲/噼啪聲
    - 關節活動時 “鎖” / “粘”
    - 關節運動範圍 (主動, 輔助, 被動)
    - 關節不穩定伴肌肉無力 / 屈曲
  - 影像學檢查: x光



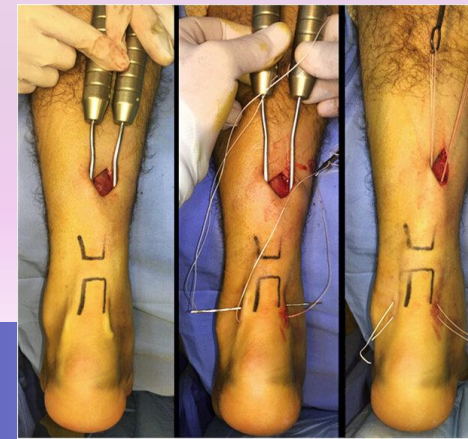
# Achilles Tendon Injury 跟腱損傷

## Surgical vs non-surgical:

- Surgical repair:
  - Reduce risk of future re-rupture
  - Both traditional open repair & minimally invasive
  - Post-surgical complications
  - Recommend for younger population, return to sports
- Non-surgical approach:
  - Functional rehabilitation approach
  - Recommend for elderly patients, less active patients

## 手術 vs 非手術：

- 手術修復：
  - 減少未來重新撕裂的風
  - 包括傳統開放修復和微創手術
  - 術後併發症
  - 建議年輕人進行，便於回到運動
- 非手術方法：
  - 功能性康復方法
  - 建議年長患者及活動量較少的患者使用



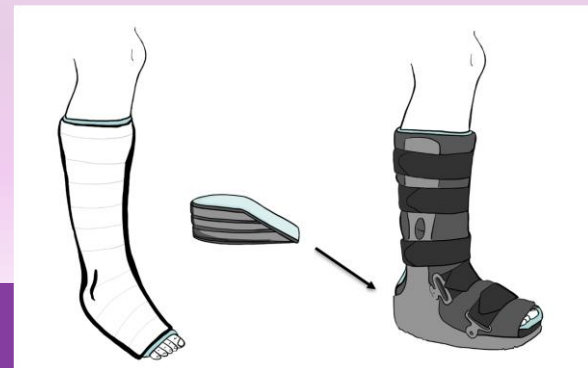
# Achilles Tendon Rupture 跟腱斷裂

## Physiotherapy Management:

- Early rehabilitation follow POLICE protocol in inflammatory phase
- Functional bracing / casting in plantar flexed position for 8-12 weeks
- Protected weight-bearing with elbow crutches
- Mobilization exercise for ROM for early phase, avoid stretching of Achilles Tendon at early phase
- Strengthening exercise for muscle strength at later stage (typically 6 weeks later)
- Proprioception training
- Functional activity training

## 物理治療管理：

- 早期康復遵循 POLICE 協議，在炎症階段進行
- 在足底屈曲位置進行功能性支具/石膏固定 8-12 週
- 使用肘拐杖進行保護性負重
- 早期階段進行活動範圍的動員運動，避免在早期階段拉伸跟腱
- 在後期（通常 6 週後）進行增強肌肉力量的訓練
- 本體感覺訓練
- 功能性活動訓練



# *Muscle Contusion*

## 肌肉挫傷

# Muscle Contusion 肌肉挫傷

- Common injuries but overall prevalence is not available due to unreported cases
- Direct, blunt blow to muscles, causing muscle bruise
- With blood vessels damage
- Common injury area:
  - Quadriceps muscles
  - Calf muscles
  - Shoulder, esp deltoid
  - Upper arm
- 常見傷害，但由於未報告的病例，整體流行率無法確定
- 直接的鈍擊對肌肉造成損傷，導致肌肉挫傷
- 伴隨血管損傷
- 常見受傷部位：
  - 股四頭肌
  - 小腿肌肉
  - 肩部，特別是三角肌
  - 上臂

# Muscle Contusion 肌肉挫傷

- Causes:
  - Contact sports, e.g. rugby, football
  - Accidents & falls
  - Impact from equipment
- Risk factors:
  - People who play contact or extreme sports
    - Male: female = 2:1
  - Age  $\geq 65$  (risk of fall)
  - Malnutrition (risk of fall)
  - Bleeding disorders, e.g. haemophilia
  - Smoking (slow healing)

原因：

- 接觸性運動，例如橄欖球足球
- 意外與跌倒
- 器材撞擊

風險因素：

- 參加接觸性或極限運動的人
- 男性：女性 = 2:1
- 年齡  $\geq 65$  歲（跌倒風險）
- 營養不良（跌倒風險）
- 出血性疾病，例如血友病
- 吸煙（癒合緩慢）

# Metro Sports Physiotherapy - 如何處理「炒芥蘭」帶來的傷患

<https://youtu.be/ddyT7XIyeo4>





# Diagnosis 診斷

- Subjective examination - onset, pattern, means to relieve
  - Objective examination:
    - Acute inflammation:
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    - 關節不穩定伴肌肉無力 / 屈曲
  - 影像學檢查: x光



# Muscle Contusion 肌肉挫傷

## Signs & symptoms:

- Varies depending on severity
- Swelling
- Skin bruising
- Weakness or stiffness of injured muscle
- Difficulty in joint movements
- Muscle haematoma

## Complications:

- Compartment syndrome
- Myositis ossificans

## 徵兆與症狀：

- 根據嚴重程度而異
- 腫脹
- 皮膚瘀傷
- 受傷肌肉無力或僵硬
- 關節運動困難
- 肌肉血腫

## 併發症：

- 腔室綜合症
- 骨化性肌炎

# Muscle Contusion 肌肉挫傷

## Medical Management:

- Drug treatment:
  - NSAIDs for short-term
  - Radiographic imaging after 3-4 weeks if still painful / cannot perform full ROM – to detect myositis ossificans

## 醫療管理：

### 藥物治療：

- 短期使用非類固醇抗炎藥（NSAIDs）
- 如果在 3-4 週後仍然疼痛或無法進行完整的活動範圍，則進行影像檢查，以檢測骨化性肌炎。

# Muscle Contusion 肌肉挫傷

## Physiotherapy Management:

- Early rehabilitation follow POLICE protocol in inflammatory phase (first 24-48 hours)
- Mobilization exercise for ROM
- Strengthening exercise for muscle strength once full & pain-free ROM achieved
- Functional activity training
- Sport-specific training

## 物理治療管理：

- 早期康復遵循 POLICE 協議，在炎症階段（前 24-48 小時）
- 進行活動範圍的動員運動
- 一旦達到完整且無痛的活動範圍，開始增強肌肉力量的訓練
- 功能性活動訓練
- 運動特定訓練

# Repetitive Strain Injury (RSI) 重複性勞損

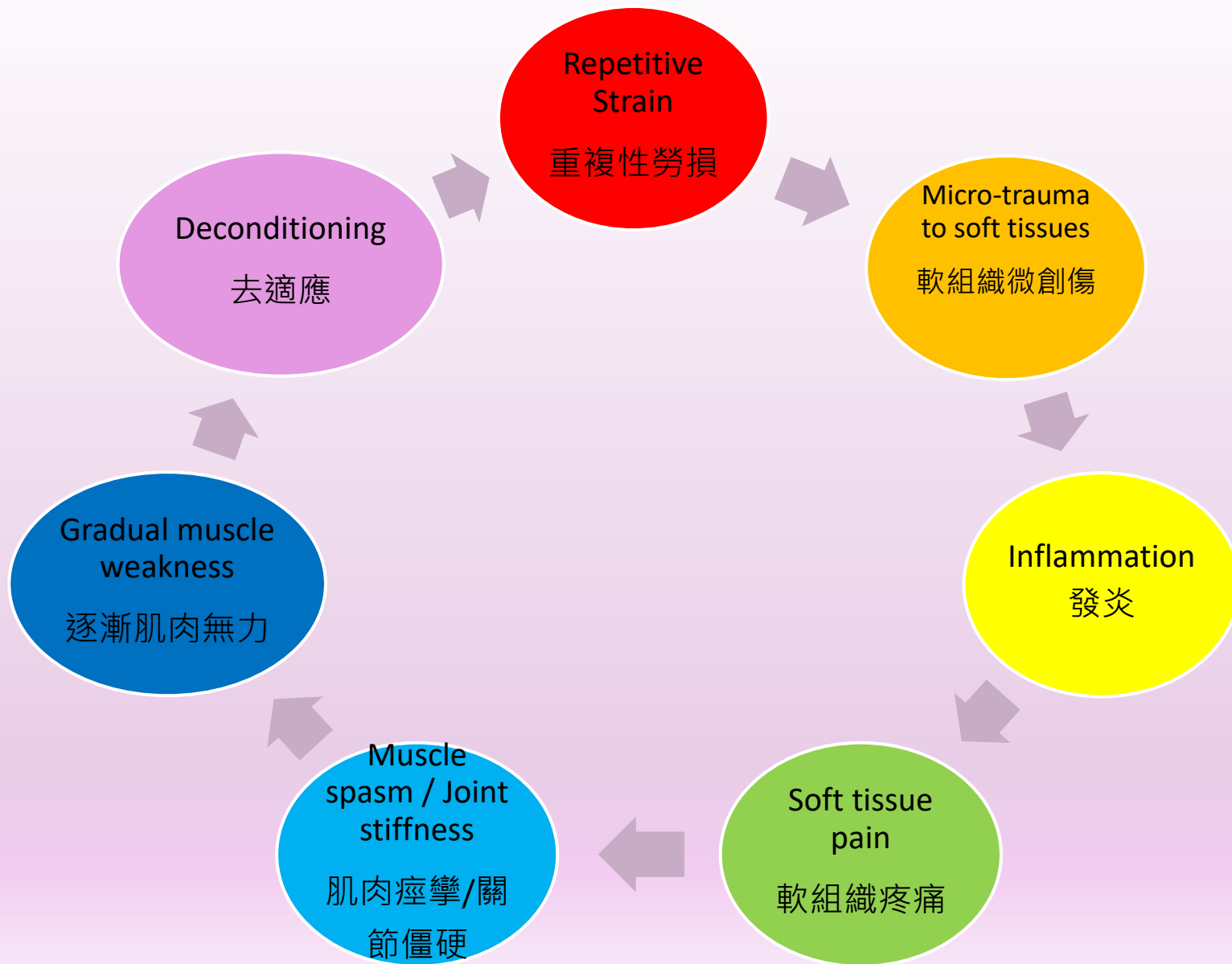
# Repetitive Strain Injury (RSI)

## 重複性勞損

- Common occupational health problem
- Gradual onset of discomfort / pain
- Affecting muscles, tendons, ligaments, nerve, bursa, etc.
- Caused by repeated micro-trauma over a period of time, without any single trauma:
  - Repetitive movements
  - Sustained postures
  - Forceful movements
  - Stress and undesirable working environment
- 常見的職業健康問題
- 不適或疼痛逐漸出現
- 影響肌肉、肌腱、韌帶、神經、滑囊等
- 由於長時間的重複微創傷所致，並非單一創傷：
  - 重複性動作
  - 持續姿勢
  - 強力動作
  - 壓力和不良工作環境

# Vicious cycle of Repetitive Strain Injury

## 重複性勞損的惡性循環





# Common Examples of RSI

## RSI 的常見範例

### Fingers & thumbs

- Trigger finger

手指與拇指

- 彈指

### Wrist

- Carpal Tunnel Syndrome

手腕

- 腕管綜合症

### Elbow

- Tennis Elbow, Golfer's Elbow

肘部

- 網球肘、golfer's elbow (高爾夫球肘)

### Shoulders

- Rotator Cuff Tendinitis

肩膀

- 旋轉袖肌腱炎

### Neck

- Tension Neck Syndrome

頸部

- 緊張性頸部綜合症

### Back

- Low back pain

背部

- 下背痛

職業勞損頸梗膊痛置之不理後果嚴重，什麼動作可預防滑鼠手、網球肘、媽媽手等？(Fit 開有條路)

<https://www.youtube.com/watch?v=GWcOeEK-6z4>



腰患、膝蓋勞損、筋膜炎等下肢的職業勞損，都是香港人工時長，長期保持一個動作或姿態引至。物理治療師教你預防方法並示範正確搬運姿勢。(Fit 開有條路)

<https://www.youtube.com/watch?v=IOjGeYduYs0>



# Common Signs & Symptoms of RSI

## RSI 的常見體徵和症狀

### Early symptoms include

- Soreness, pain, tingling & discomfort when performing the repetitive task
- Symptoms reduced when stop the task

### Common signs & symptoms if left untreated and gradual deteriorate:

- Burning, aching or shooting pain
- Swelling
- Stiffness
- Throbbing
- Tingling / numbness
- Muscle weakness
- Muscle cramp

### 早期症狀包括：

- 在進行重複性任務時出現酸痛、疼痛、刺痛和不適
- 停止任務後，症狀減輕

### 如果未經治療，常見的徵兆與症狀會逐漸惡化：

- 燒灼感、疼痛或刺痛
- 腫脹
- 僵硬
- 跳動感
- 刺痛 / 麻木
- 肌肉無力
- 肌肉痙攣

# General Risk Factors for RSI

## RSI 的一般風險因素

- Work posture (esp. awkward or improper posture)
  - work at a non-ergonomically designed workspace
- Heavy physical work
- Lifting
- Repetitive, labour-intensive work
  - prolonged use on computer, keyboard & mouse
- Vibration (via hand tools)
- Extreme low temperature
- Individual risk factors:
  - Smoking
  - High Body Mass Index (BMI)
  - Co-morbidities
- 工作姿勢 (尤其是尷尬或不當的姿勢)
  - 在非人體工學設計的工作空間工作
- 重體力工作
- 提舉
- 重複性、勞動密集型工作
  - 長時間使用電腦、鍵盤和鼠標
- 震動 (通過手工具)
- 極低溫度
- 個人風險因素：
  - 吸煙
  - 高體重指數 ( BMI )
  - 併發症



# Carpal Tunnel Syndrome 腕管綜合症

## What is Carpal Tunnel Syndrome?

- Compression of median nerve at the wrist as it passes through the carpal tunnel

## Carpal Tunnel:

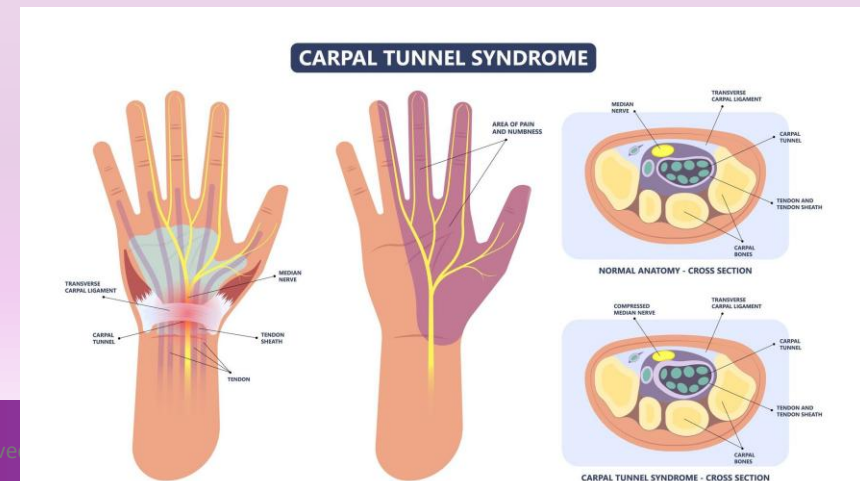
- U-shaped canal consists of carpal bones, with a dense fibrous tissue (Transverse Carpal Ligament) covering on top
- Median Nerve & Flexor Tendons of hand pass through this tunnel

## 什麼是腕管綜合症？ 腕管綜合症是指在手腕處

- 正中神經受壓迫的情況，當它通過腕管時會受到影響。

## 腕管：

- U形的通道，由腕骨組成，上方覆蓋著密集的纖維組織（橫腕韌帶）。
- 正中神經和手部屈肌腱通過此通道。



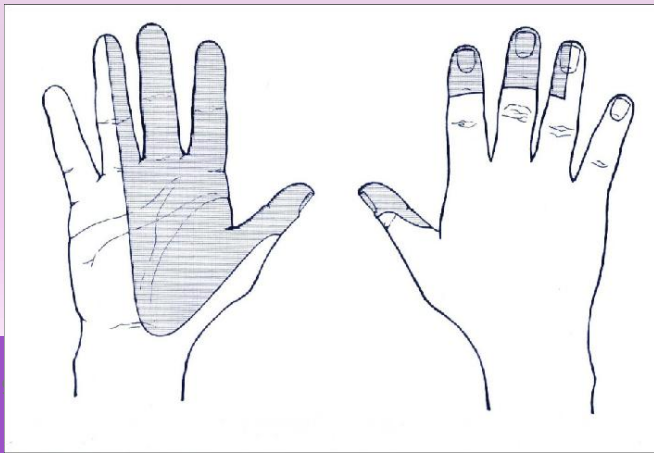
# Median Nerve 正中神經

## Skin Sensation:

- Lateral aspect of palm
- Lateral three digits
- Lateral half of the 4<sup>th</sup> digit
- Dorsal aspects of the distal halves of these digits

## Motor innervation:

- Thenar muscles (movements of thumb)

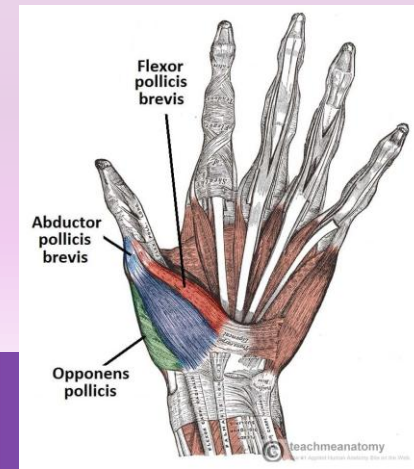


## 皮膚感覺：

- 手掌外側
- 外側三個手指
- 第四個手指的外側一半
- 這些手指遠端一半的背側

## 運動神經支配：

- 大魚際肌（ 拇指的運動 ）





# Carpal Tunnel Syndrome 腕管綜合症

**Compression on the median nerve at the wrist**

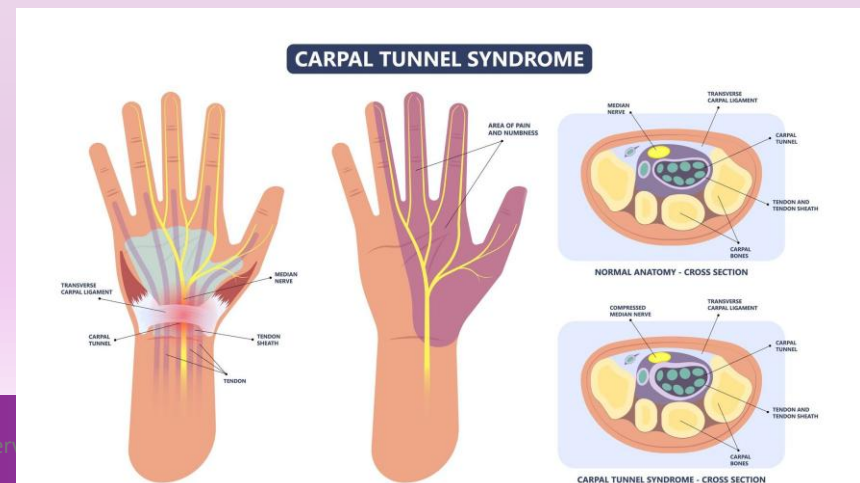
## **Signs & Symptoms:**

- Pain, numbness, tingling of first THREE and half fingers
- Muscles weakness at hand with atrophy at thenar eminence
- Wrist pain / numbness that disturb sleep

**Can get worse over time, early diagnosis & treatment is necessary**

**正中神經在手腕處受到壓迫  
徵兆與症狀：**

- 前三個手指及一半手指的疼痛、麻木和刺痛
- 手部肌肉無力，並伴隨大魚際肌萎縮
- 手腕疼痛/麻木，影響睡眠
- **症狀可能隨著時間惡化，因此及早診斷和治療是必要的**



# Risk Factors of Carpal Tunnel Syndrome

## 腕管綜合症的風險因素

- Repeated motions at wrist contribute swelling & compression of median nerve
- Diabetes mellitus (DM)
- Fluid retention from pregnancy / menopause
- Autoimmune conditions such as rheumatoid arthritis
- Fractures/ trauma to wrist
- Female / individuals with smaller carpal tunnel
- Obesity
- 手腕的重複性動作會導致腫脹和正中神經的壓迫
- 糖尿病
- 懷孕/更年期引起的液體滯留
- 自體免疫疾病，如類風濕性關節炎
- 手腕骨折/創傷
- 女性/手腕較小的人
- 肥胖

# Diagnosis 診斷

- Subjective examination - onset, pattern, means to relieve
- Objective examination:
  - Acute inflammation: Redness, joint swelling, warmth & pain
  - Range of motion (active, assisted, passive)
  - Muscle strength
  - Special tests: Tinel's sign, Phalen's Test, etc.
- Investigative tests:
  - Nerve Conductive Test: nerve conduction speed of median nerve
  - Ultrasound imaging
  - Magnetic Resonance Imaging (MRI)
- 主觀檢查 - 發病時間、模式、緩解方法
- 客觀檢查：
  - 急性炎症：發紅、關節腫脹、熱感及疼痛
  - 活動範圍（主動、輔助、被動）
  - 肌肉力量
  - 特殊測試：如 Tinel 徵象、Phalen 測試等
- 檢查測試：
  - 神經傳導測試：測量正中神經的神經傳導速度
  - 超聲影像
  - 磁共振成像（MRI）



# Management of CTS

## Physiotherapy Management

- Stretching Exercises: Wrist flexor and extensor stretches to improve flexibility.
- Wrist Splinting: Use splints to keep the wrist in a neutral position, especially at night, to avoid excessive flexion.

## Medical Management

- Medications: NSAIDs to relieve pain and inflammation.
- Steroid Injections: Injections in the carpal tunnel to reduce inflammation and swelling.
- Treat Underlying Conditions: Manage diabetes and arthritis to improve symptoms.

## 物理治療管理

- 伸展運動: 手腕屈肌和伸肌的伸展運動以改善靈活性。
- 手腕護具: 使用護具保持手腕在中立位置，尤其在夜間，避免過度屈曲。

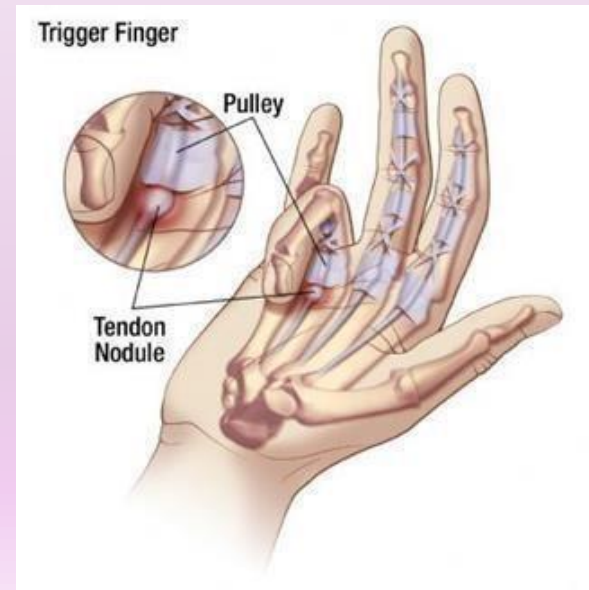
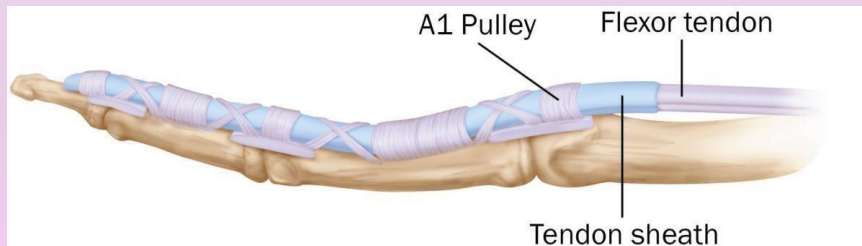
## 醫療管理

- 藥物治療: 使用非類固醇消炎藥 ( NSAIDs ) 來緩解疼痛和炎症。
- 類固醇注射: 在腕管內進行注射以減少炎症和腫脹。
- 治療潛在疾病: 管理糖尿病和關節炎以改善症狀。



# Trigger Finger 板機指

- Inflammation narrows the space within the sheath that surrounds the tendon in the affected finger
- Affects ONE/ MORE of tendons at fingers, difficult to bend affected finger or thumb
- 炎症會縮小包圍受影響手指肌腱的鞘內空間
- 影響一根或多根手指的肌腱，導致彎曲受影響的手指或拇指困難。



# Common Signs & Symptoms of Trigger finger

## 扳機指的常見體徵和症狀

- Finger stiffness, particularly in the morning
- A popping / clicking sensation when moving
- Tenderness / a bump (nodule) in the palm at the base of affected finger
- Finger catching / locking in a bent position, fail to straighten, that suddenly pops straight
- • +/- acute inflammation - redness, swelling, increase local temperature, pain
- 手指僵硬，特別是在早晨
- 移動時出現彈響或咔嗒聲
- 手掌基部出現觸痛或腫塊（結節）
- 手指在彎曲位置卡住/鎖住，無法伸直，突然彈直
- +/- 急性炎症：發紅、腫脹、局部溫度升高、疼痛

# Risk factors of trigger fingers

## 扳機指的風險因素

- Repeated gripping
- Health conditions-diabetes /rheumatoid arthritis
- Carpal tunnel syndrome surgery
- Gender-female more common
- 重複性握持
- 健康狀況：糖尿病 / 類風濕性關節炎
- 腕管綜合症手術
- 性別：女性更常見



# Management of trigger finger 扳機指的管理

## Physiotherapy Management

- Stretching Exercises: Finger stretches to improve flexibility and reduce stiffness.
- Manual Therapy: Techniques to mobilize the affected finger and reduce tension around the tendons.
- Splinting: Use splints to immobilize the finger in a neutral position, especially at night.
- Activity Modification: Educate on avoiding repetitive gripping motions that exacerbate symptoms.

## Medical Management

- Medications: NSAIDs for pain and inflammation relief.
- Corticosteroid Injections: Injections at the affected site to reduce inflammation and swelling.
- Surgery Surgical release of the tendon sheath may be considered for severe cases not responding to conservative treatment.

## 物理治療管理

- 伸展運動: 手指伸展以改善靈活性並減少僵硬。
- 手動療法: 技術以動員受影響的手指，減少肌腱周圍的緊張感。
- 護具使用: 護具將手指固定在中立位置，特別是在夜間。
- 活動調整: 教育患者避免重複性的握持動作，以減輕症狀。

## 醫療管理

- 藥物治療: 使用非類固醇消炎藥 (NSAIDs) 來緩解疼痛和炎症。
- 類固醇注射: 在受影響部位進行注射以減少炎症和腫脹。
- 手術對於對保守治療無反應的嚴重病例，可能考慮進行肌腱鞘釋放手術。

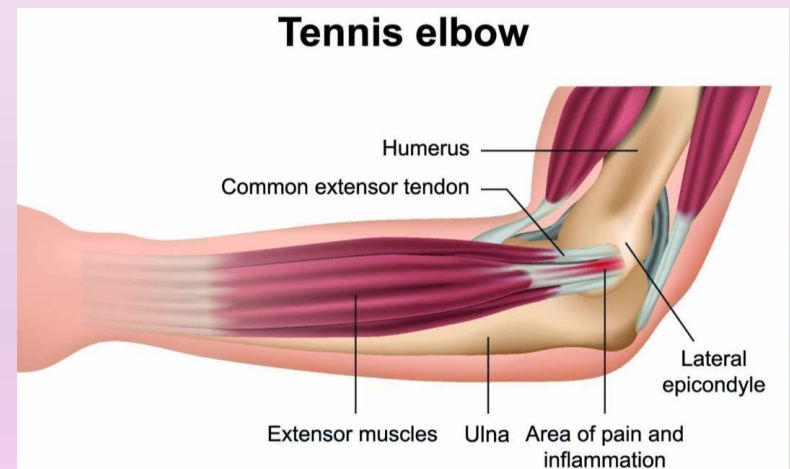
# Tennis elbow / Lateral epicondylitis

## 網球肘

- a painful condition of the elbow caused by overuse
- may result in playing tennis or other racquet sports
- an inflammation of the tendon that join the forearm muscles on
- the outside of the elbow (extensors origin)

肘部疼痛的病徵，由於過度使用所致

- 可能由於打網球或其他球拍運動引起
- 是肘部外側前臂肌肉（伸肌起點）連接的肌腱發炎



# Management of Tennis Elbow

## 網球肘的管理

### Physiotherapy Management

- Stretching Exercises: Stretches for the forearm extensors to improve flexibility.
- Strengthening Exercises: Strengthening exercises for the forearm and wrist muscles once pain subsides.
- Manual Therapy: Techniques to reduce muscle tension and improve blood circulation.
- Ice Therapy: Application of ice to reduce inflammation and pain.
- Activity Modification: Educating patients on avoiding repetitive motions that aggravate the condition.

### Medical Management

- Medications: Nonsteroidal anti-inflammatory drugs (NSAIDs) for pain and inflammation relief.
- Corticosteroid Injections: Injections at the affected site to reduce inflammation.
- Surgery: Considered for severe cases that do not respond to conservative treatment.

### 物理治療管理

- 伸展運動: 針對前臂伸肌的伸展運動以改善靈活性。
- 強化運動: 在疼痛緩解後進行的前臂和手腕肌肉加強運動。
- 手動療法: 減少肌肉緊張和改善血液循環的技術。
- 冰敷療法: 使用冰敷以減輕炎症和疼痛。
- 活動調整: 教育患者避免加重病情的重複動作。

### 醫療管理藥物治療:

- 使用非類固醇消炎藥 (NSAIDs) 來緩解疼痛和炎症。
- 類固醇注射: 在受影響部位進行注射以減少炎症。
- 手術: 對於對保守治療無反應的嚴重病例，可能考慮進行手術。

# Plantar Fasciitis 足底筋膜炎

- strong band of tissue-plantar fascia (at bottom of foot) becomes irritated & inflamed.
- Fascia for absorb high stress & strains

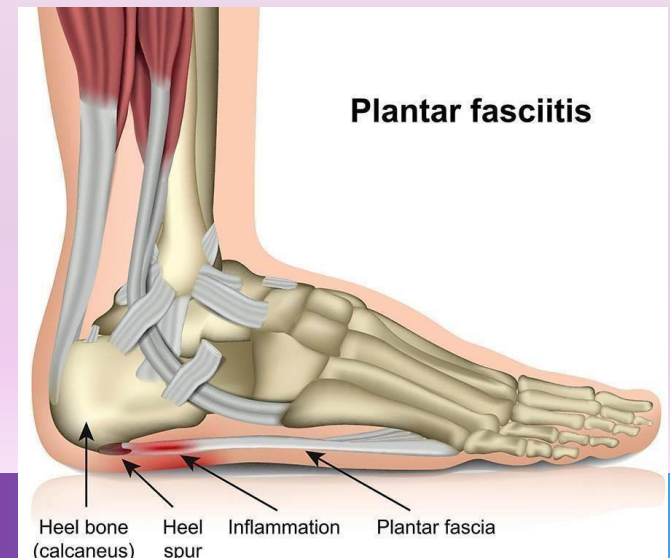
## Causes

- Too much stress & strains
- result in heel pain & stiffness at fascia

- 強韌的組織帶 - 足底筋膜 (位於腳底) 變得刺激和發炎
- 筋膜用於吸收高強度的壓力和拉力。

## 原因

- 過多的壓力和拉力
  - 導致腳跟疼痛和筋膜僵硬。



# Risk factor of Plantar Fasciitis

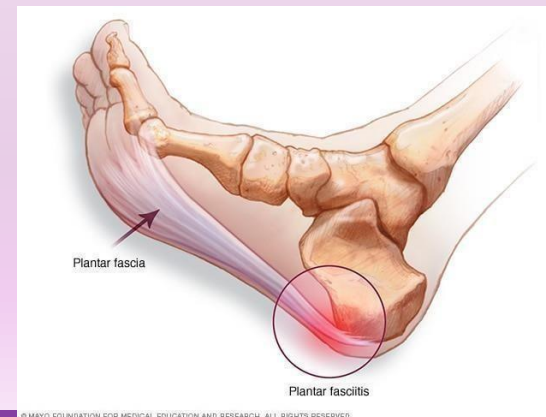
## 足底筋膜炎的風險因素

- Tightness at calf muscles
- Obesity
- Very high foot arch
- Repetitive impact activities (running/sports)
- New/ increased activity
- 小腿肌肉緊繃
- 肥胖
- 非常高的足弓
- 重複性衝擊活動（如跑步/運動）
- 新活動或增加活動量

# Signs & Symptoms of Plantar Fasciitis

## 足底筋膜炎的體徵和症狀

- Pain at the bottom of foot near heel
- Pain with FIRST few steps after get out from bed/ after long period of sit. Usually subside after few minute of walk
- Greater pain felt after exercise
- +/- Acute inflammation
- 腳底靠近腳跟的疼痛
- 在床上起床後或長時間坐下後的前幾步會感到疼痛，通常在走幾分鐘後會減輕
- 運動後感到更劇烈的疼痛
- 可能伴隨急性炎症



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# Management of Plantar Fasciitis

## 足底筋膜炎的管理

### Physiotherapy Management

- Stretching Exercises: Focus on stretching the calf muscles and plantar fascia to improve flexibility.
- Strengthening Exercises: Strengthening exercises for the foot and lower leg muscles to support the arch.
- Manual Therapy: Techniques to relieve tension in the fascia and improve range of motion.
- Ice Therapy: Application of ice to reduce inflammation and pain.
- Footwear Assessment: Guidance on appropriate footwear to provide proper support.
- Orthotics: Custom or over-the-counter orthotic devices to support the arch and relieve pressure.
- Activity Modification: Educating patients on avoiding activities that exacerbate symptoms.

### Medical Management

- Medications: Nonsteroidal anti-inflammatory drugs (NSAIDs) to relieve pain and inflammation.
- Corticosteroid Injections: Injections at the affected site to reduce inflammation.
- Surgery: Considered for severe cases that do not respond to conservative treatments.

### 物理治療管理

- 伸展運動: 針對小腿肌肉和足底筋膜的伸展運動以改善靈活性。
- 強化運動: 加強足部和小腿肌肉的運動以支持足弓。
- 手動療法: 減少筋膜緊張和改善活動範圍的技術。
- 冰敷療法: 使用冰敷以減輕炎症和疼痛。
- 鞋類評估: 提供適合的鞋類建議以提供適當支持。
- 矯形器具: 定制或市售的矯形器具以支持足弓並減輕壓力。
- 活動調整: 教育患者避免加重症狀的活動。

### 醫療管理

- 藥物治療: 使用非類固醇消炎藥 (NSAIDs) 來緩解疼痛和炎症。
- 類固醇注射: 在受影響部位進行注射以減少炎症。
- 手術: 對於對保守治療無反應的嚴重病例，可能考慮進行手術。



# E-Resources

- <https://www.youtube.com/watch?v=LjVe16W56ml>
- [https://www.youtube.com/watch?v=yMsVe4HECxs&list=PLNAXUDE\\_mTeEQyGuKxMROm5FWqVenCMAF&index=12](https://www.youtube.com/watch?v=yMsVe4HECxs&list=PLNAXUDE_mTeEQyGuKxMROm5FWqVenCMAF&index=12)

## 使用智能手機或電腦的職安健要點

- <https://www.youtube.com/watch?v=j83gkOJkJMc>

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**Prepared by: Ms Evelyn KO, Lecturer, RPT**

**Updated by: Mr Wollo WU, Senior Lecturer, RPT  
Ms Edith TAM, Lecturer, RPT  
Ms Bonnie MA, Lecturer, RPT**